Infineon/BIT Excellence Program 2026

Sustainable Supply Chain for Dried Mango from Africa to Germany

Module 08
Machinery Management System

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1 Machinery Management Large Scope Definition and Equipment Registry

1.1 Scope and Perimeter of Work

The Machinery Management System is designed to track and manage every piece of mechanized or hand-operated equipment used across the farm's production chain, from mango cultivation to the processing of dried mango. Rather than treating equipment as a side note within each production stage, this module gathers all physical assets into a single registry so that their maintenance, usage, and lifecycle can be monitored independently of the process they support.

In scope are all equipment items directly involved in growing, harvesting, and processing mangoes: agricultural machinery such as the tractor or power tiller and the motor pump used for irrigation, hand tools used for planting and harvesting, post harvest handling equipment, and the full processing line made up of the elevators, washer, sorter, cleaner, dryer, and peeler. For each of these assets, the system covers six areas: maintenance scheduling, usage and operating hours, repair history, fuel consumption, operator assignment, and spare parts inventory.

Out of scope, for now, are the farm's energy generation assets, namely the solar panels and batteries covered under the Energy Supply Systems module, and the vehicles used to transport dried mango from Banfora to Germany, which are assumed to fall under the Sales and Marketing module's delivery tracking. These two boundaries still need to be confirmed with the owners of the modules concerned.

A deliberate distinction is made between owning a process and owning an asset. The Water Supply module remains responsible for irrigation scheduling and water quality, and the Product Transformation module remains responsible for drying quality and food safety compliance. The Machinery module, by contrast, is only responsible for the physical equipment itself, such as the motor pump or the dryer, and for keeping that equipment operational. This separation avoids duplicating data between modules while still giving Machinery full visibility over every asset the farm depends on.

1.2 Data Consolidation from 2025 Cohort

Before defining this year's equipment list, the 2025 cohort's Excel file was reviewed sheet by sheet to identify any existing data that could serve as a starting point, in line with the instruction to study last year's sustainable farm and use it as a foundation.

On Zalka's Plants sheet, Table 4 (Equipment Pricing) lists five items: tractor and power tiller, motor pump and irrigation, planting and harvesting tools, post harvest equipment, and inputs such as fertilizers and seeds. The total shown, 15,548 EUR, is the sum of the first four lines only, correctly leaving inputs out of the equipment total. Table 6 (Purchase Price of Plots), however, later lists an equipment line of 19,362 EUR, which corresponds to 15,548 EUR plus the 3,814 EUR of inputs added back in. This inconsistency, inputs being excluded from equipment in one table and included in another, was one of the first issues identified in the source file. It confirms that inputs such as fertilizers and seeds should be treated as operational costs rather than equipment throughout this module.

The same sheet's bibliography section also clarifies what each rolled up line actually contains. The tractor and power tiller line and the motor pump and irrigation line correspond respectively to a mini tractor priced at 6,860 EUR and a motor pump with hose kit priced at 4,573 EUR. The planting and harvesting tools line, at 1,829 EUR, covers a steel hoe, a branch cutter, pruning shears, nursery trays, a manual row transplanter, a wheelbarrow, and a thermal motorized sprayer. The post harvest equipment line, at 2,286 EUR, covers a sorting table, a brushing unit, a belt conveyor, a packing scale, an impulse sealer, a manual pallet jack, and plastic crates. Once these itemized lists were matched against the rolled up totals, none of the original categories turned out to be arbitrary, they had simply been summarized without their supporting detail.

Cedric's Smart Dryer sheet contributed the processing side of the equipment list: elevators, bubble washer, sorter, cleaner, dryer, and peeler, priced individually and totaling 8,861 EUR. These items were kept in a separate sheet from Zalka's list in last year's work, even though they belong to the same physical value chain. This year, they have been merged into a single registry rather than split by production stage.

Abdoul's work on the solar PV power plant was reviewed as well, but the battery and solar tracking data it contains describes energy generation assets rather than mechanized farm equipment. It has therefore been left out of this module and is expected to remain under Energy Supply Systems.

The logic applied to this year's Machinery module follows directly from these observations: every physical, mechanized or hand-operated asset used in cultivation, harvest, or processing is tracked once, under a single asset registry, regardless of which stage of production it supports, while consumables such as fertilizers, seeds, fuel, and packaging materials, along with energy generation assets, are explicitly kept out.

1.3 Consolidated Equipment List

The table below merges the equipment data scattered across last year's Plants and Smart Dryer sheets into a single list, with fictional asset IDs assigned for use in this year's prototype. Unit costs are carried over from the 2025 figures, since they still represent the best available cost estimate for a 2-hectare farm and allow this year's work to build directly on last year's business case.

Asset IDs follow the pattern MCH-XXX and are entirely fictional at this stage, meant to populate the frontend prototype rather than to reflect real serial numbers or tags. Once the farm is operational, each ID would be replaced by a physical asset tag linked to the equipment's actual purchase record.

1.4 Data Model

The data model below turns the six focus areas of this module, namely maintenance, usage, repairs, fuel, operators, and spare parts, into eleven linked entities grouped into four management families, all built around the Equipment table as the central reference. Every other entity refers back to an asset through its `asset_id`, which keeps the model consistent and avoids repeating equipment details in each log. This structure is what will be used to generate the fictional dataset behind the frontend prototype.

Figure 2 lists the eleven entities grouped by family, while Figure 3 shows how they relate to one another.

Asset ID	Equipment	Category	Stage	Source
MCH-001	Tractor / power tiller	Agricultural machinery	Cultivation	Zalka
MCH-002	Motor pump (irrigation asset)	Agricultural machinery	Cultivation	Zalka
MCH-003	Planting & harvesting tool set	Agricultural equipment	Cultivation	Zalka
MCH-004	Post-harvest handling equipment	Agricultural equipment	Harvest	Zalka
MCH-005	Elevators	Processing machinery	Processing	Cedric
MCH-006	Bubble washer	Processing machinery	Processing	Cedric
MCH-007	Sorter	Processing machinery	Processing	Cedric
MCH-008	Cleaner	Processing machinery	Processing	Cedric
MCH-009	Dryer	Processing machinery	Drying	Cedric
MCH-010	Peeler	Processing machinery	Processing	Cedric

Figure 1: Consolidated equipment list with fictional asset IDs for prototype development.

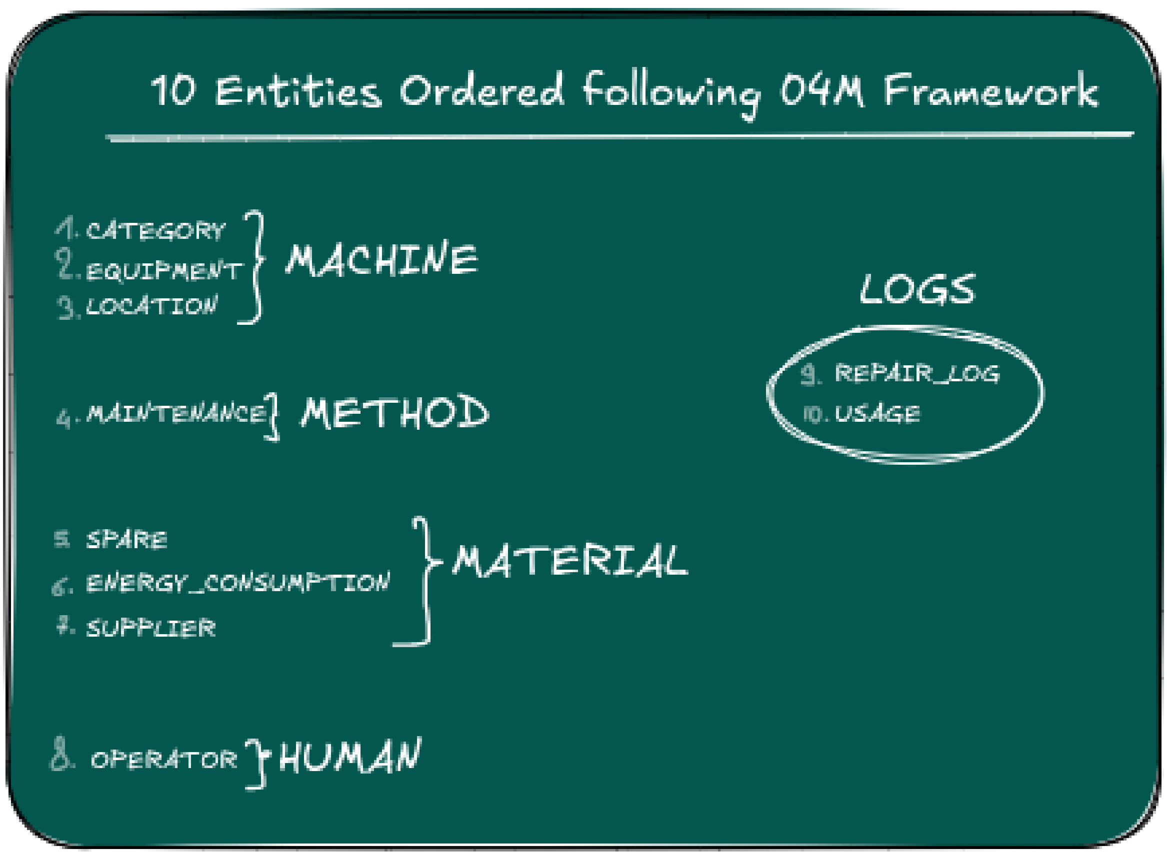

Figure 2: Entity list of the Machinery data model, grouped by management family.

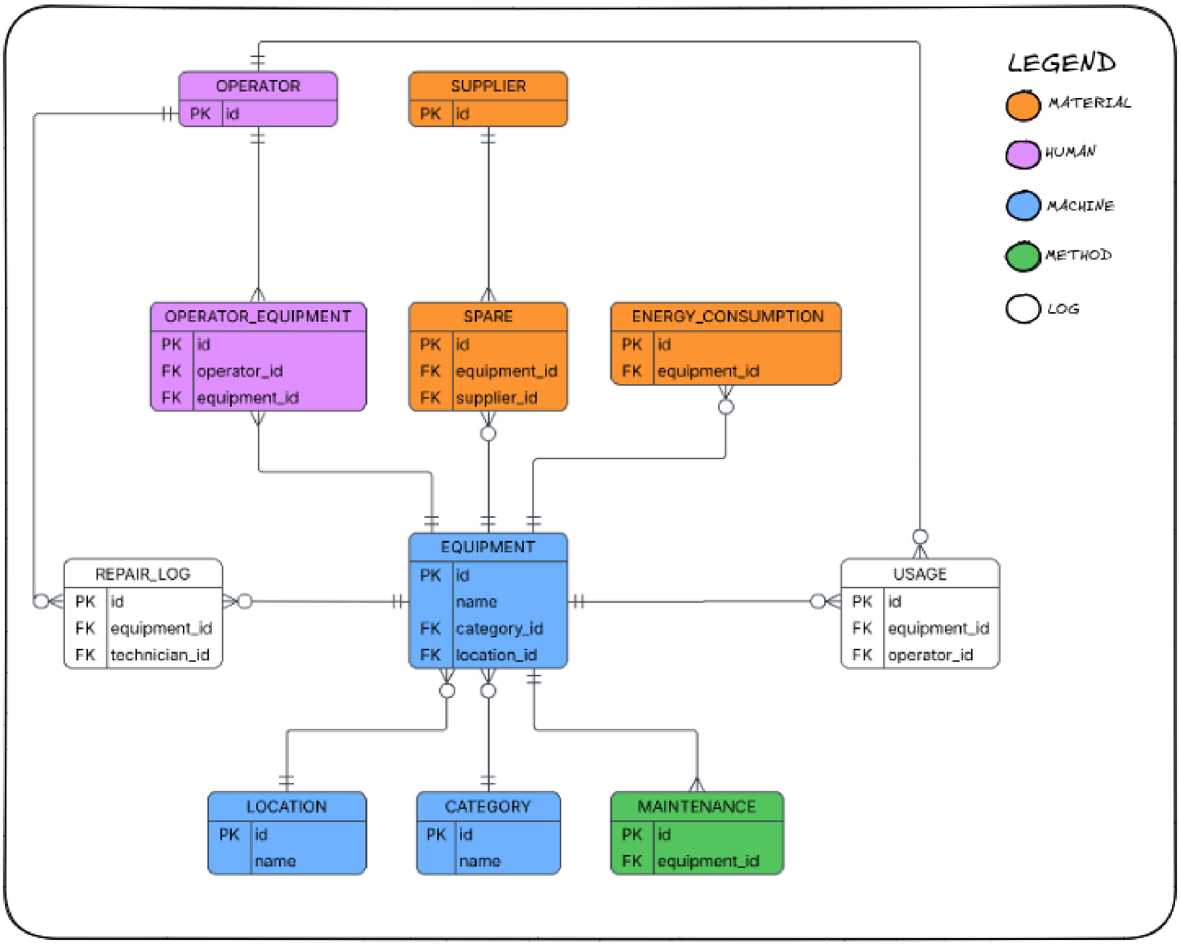

Figure 3: Entity relationship diagram of the Machinery data model.

Entity	Description	Example
Machine		
Equipment	Central asset registry; PK `asset_id`.	MCH-001 – Tractor / power tiller
Category	Classification of each asset.	Agricultural machinery / Processing machinery
Location	Site where the asset is currently based.	Zalka / Cedric
Method		
Maintenance	Preventive / corrective maintenance planning per asset.	MCH-009 Dryer, every 250 h, next due 15/07/2026
Material		
Spare	Spare parts stock with reorder threshold.	Filter for MCH-009, qty 4, threshold 2
Supplier	Suppliers of parts and equipment.	AgriParts SARL
Energy_Consumption	Fuel / energy consumption per asset.	MCH-001, 45 L, 120 EUR
Human		
Operator	Registry of operators.	Omar El Amrani, Driver
Operator_Equipment	Assignment of an operator to an asset.	Omar → MCH-001 since 01/02/2026
Transversal Logs (Machine + Human)		
Usage	Usage log: date and hours operated.	MCH-001, 12/05/2026, 6.5 h
Repair_Log	Repair history: fault, downtime, cost.	MCH-005, conveyor belt, 8 h, 350 EUR

Table 1: Overview of the eleven entities with their description and an example record.

1.5 Frontend Prototype

A working frontend mockup of the Machinery Management System is available and linked from my personal homepage tracker, where it is kept up to date alongside the rest of the project work. Rather than capturing a static screenshot in this report, readers are invited to check the live version there, as the interface continues to evolve together with the underlying data model.

1.6 Open Questions on the Technical Stack

With the mockup validated by the mentor, the next milestone is choosing the technology stack for the backend implementation. Anar, brought onto the weekly call as computer science support, will be available to answer these questions. The following points were prepared ahead of the call to keep the discussion short and focused.

Technical Domain	Open Question
Frontend	Which stack should be used for the frontend?
Backend	Which language and framework should be used for the backend API?
Database	Which database solution should be chosen for this project?
Deployment Target	Should this be built as a web application or as a desktop application?
Version Control	Will a shared repository be provided for all modules, or should each module use its own separate GitHub repository?

Table 2: Open questions to be addressed during technical stack selection.